

Function Lainnya: LAG, LEAD, RUNNING TOTAL (SUM() OVER), MOVING AVERAGE (AVG() OVER)

Dalam hal ini, kita gunakan database Sales FMCG 2022-2024.

1. LAG

Fungsi: Mengambil nilai dari **baris sebelumnya** tanpa perlu melakukan JOIN ke tabel yang sama.

LAG(nama_kolom, offset, default_value)

OVER(...)

Tujuan pembelajaran: Membandingkan Penjualan dengan Bulan Sebelumnya

SELECT

sku,

sales_year,

sales_month,

units_sold,

LAG(units_sold)

OVER(

PARTITION BY sku

ORDER BY sales_year, sales_month

) AS Previous_Sales,

units_sold -

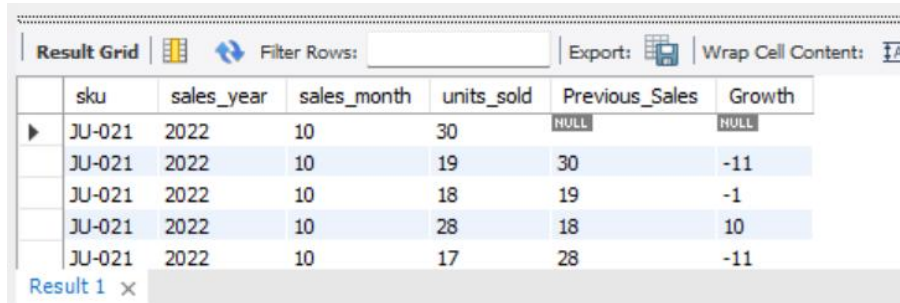
LAG(units_sold)

OVER(PARTITION BY sku

ORDER BY sales_year, sales_month

) AS Growth

FROM fmcg_clean_data;



	sku	sales_year	sales_month	units_sold	Previous_Sales	Growth
▶	JU-021	2022	10	30	NULL	NULL
	JU-021	2022	10	19	30	-11
	JU-021	2022	10	18	19	-1
	JU-021	2022	10	28	18	10
	JU-021	2022	10	17	28	-11

2. LEAD

Tujuan pembelajaran: Melihat penjualan berikutnya

SELECT

sku,

sales_month,

units_sold,

LEAD(units_sold)

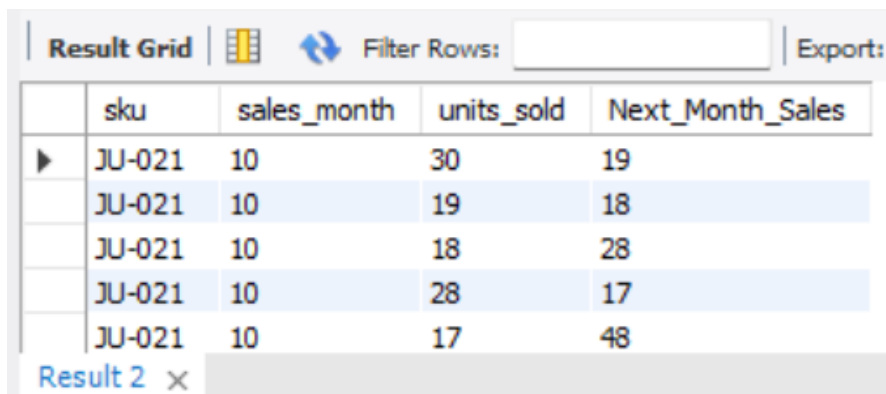
OVER(

PARTITION BY sku

ORDER BY sales_year, sales_month

) AS Next_Month_Sales

FROM fmcg_clean_data;



	sku	sales_month	units_sold	Next_Month_Sales
▶	JU-021	10	30	19
	JU-021	10	19	18
	JU-021	10	18	28
	JU-021	10	28	17
	JU-021	10	17	48

3. RUNNING TOTAL

Tujuan pembelajaran: Misalnya ingin mengetahui total penjualan kumulatif setiap SKU.

```
SELECT
```

```
    sku,
```

```
    sales_year,
```

```
    sales_month,
```

```
    units_sold,
```

```
    SUM(units_sold)
```

```
    OVER(
```

```
        PARTITION BY sku
```

```
        ORDER BY sales_year, sales_month desc
```

```
    ) AS Running_Total
```

```
FROM fmcg_clean_data;
```

	sku	sales_year	sales_month	units_sold	Running_Total
	JU-021	2022	8	11	9974
	JU-021	2022	8	4	9974
	JU-021	2022	8	9	9974
	JU-021	2022	8	35	9974
	JU-021	2022	8	10	9974

Result 4 ×

4. MOVING TOTAL

Tujuan Pembelajaran: Misalnya ingin mengetahui rata-rata penjualan **3 bulan terakhir**.

```
WITH monthly_sales AS (
```

```
    SELECT
```

```
        sku,
```

```
        sales_year,
```

```
    sales_month,  
    SUM(units_sold) AS units_sold  
FROM fmcg_clean_data  
GROUP BY  
    sku,  
    sales_year,  
    sales_month  
)
```

```
SELECT  
    sku,  
    sales_year,  
    sales_month,  
    units_sold,  
    ROUND(  
        AVG(units_sold) OVER(  
            PARTITION BY sku  
            ORDER BY sales_year, sales_month  
            ROWS BETWEEN 2 PRECEDING AND CURRENT ROW  
        ),  
        2  
    ) AS Moving_Average_3M  
FROM monthly_sales  
ORDER BY  
    sku,  
    sales_year,  
    sales_month;
```

Result Grid

Filter Rows:

Export:



Wrap Cell Content:

	sku	sales_year	sales_month	units_sold	Moving_Average_3M
▶	JU-021	2022	10	4988	4988.00
	JU-021	2022	11	4512	4750.00
	JU-021	2022	12	4364	4621.33
	JU-021	2022	7	3469	4115.00
	JU-021	2022	8	4510	4114.33